

From first principles to optimised readout of a SHiP timing bar

Photon transport, order-statistic estimators, and BLUE combination

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Geometry of a single bar

- Scintillator: $1400 \times 60 \times 10 \text{ mm}^3$
- END readout: **8 SiPMs/end** $\times$ **2 ends** = **16 total**
- TOP readout (hybrid): **20 SiPMs** sparse along top face
- Hybrid total:
 $N_{\text{END}}=16 + N_{\text{TOP}}=20 = N_{\text{total}}=36$

Detector scale

- Bar area $\approx 1.4 \times 0.06 = 0.084 \text{ m}^2$
- Full detector $\approx 50 \text{ m}^2 \Rightarrow \sim 600$ bars
- $N_{\text{total}} = 36 \Rightarrow \sim 21\,600$ channels total
- Each saved SiPM per bar removes ~ 600 channels

Design question (not merely "smallest σ "):

What is the *minimum practical* instrumentation that delivers event time AND position for each bar?

Figures of merit:

- Timing resolution σ_t
- Longitudinal position σ_x
- Response uniformity along bar
- Channel count
- Robustness to SiPM/electronics jitter

Why an analytic model?

- Builds physical intuition before simulation
- Sets order-of-magnitude expectations
- Diagnoses surprises in Geant4 output
- Identifies dominant parameters

Chain

$$\mu \xrightarrow{L} E_{\text{dep}} \rightarrow N_{\gamma} \rightarrow f_{\text{TIR}} \rightarrow P_{\text{surv}} \rightarrow N_{\text{pe}}$$

Crucial distinction

The napkin gives *scales and trends*. Agreement at $\sim 20\%$ level is a success, not a precision measurement.

This talk

- 1 First-principles photon transport
- 2 Geant4 validation / photon budget
- 3 Estimator optimisation
- 4 Position reconstruction
- 5 Detector design outlook

For a minimum-ionising particle in plastic
($\rho \approx 1.03 \text{ g/cm}^3$):

$$\left\langle \frac{dE}{dx} \right\rangle \approx 2 \text{ MeV/cm}$$

Bar thickness along muon track: $T = 1 \text{ cm}$
(vertical incidence)

$$E_{\text{dep}} \approx 2 \text{ MeV} \quad (\text{Landau peak})$$

The Landau distribution has a *tail* to high values;
the napkin uses only the most probable value.

Simulation input

Geant4 uses the full Landau distribution event-by-event. The $\sim 2 \text{ MeV}$ scale is confirmed by the simulation; event-by-event fluctuations in N_γ propagate into the timing residuals.

Practical consequence: photon number varies event-by-event (Landau + Poisson), broadening the timing distribution beyond pure photon statistics.

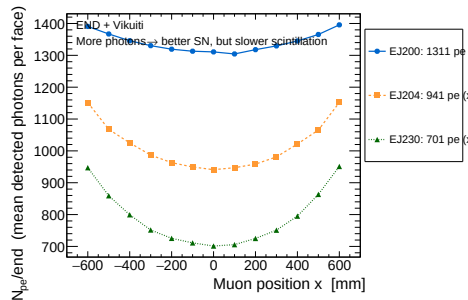
For EJ-230 (scintillation yield
 $Y = 9700$ photons/MeV):

$$N_\gamma = Y \cdot E_{\text{dep}} = 9700 \times 2 \approx 1.9 \times 10^4$$

All three materials for comparison:

Material	Yield [ph/MeV]	τ [ns]	λ_{att} [mm]
EJ-200	10 000	2.1	3800
EJ-204	10 400	1.8	1600
EJ-230	9 700	1.5	900

Napkin prediction: EJ-230 wins timing despite fewer photons, because shorter τ provides earlier peak arrival.



Geant4: N_{pe}/end vs x , END + Vikuiti. EJ-200 > EJ-204 > EJ-230 in photon count — matches napkin.

Critical angle for plastic–air interface ($n = 1.58$):

$$\theta_c = \arcsin\left(\frac{n_{\text{air}}}{n}\right) \approx 39.3^\circ$$

For one transverse wall family, fraction with $|\cos \theta_\perp| < \cos \theta_c$:

$$a = \cos \theta_c \approx 0.774 \quad (\text{single wall})$$

For the *rectangular bar* (two independent transverse families):

$$f_{\text{TIR}} = 2a - 1 \approx 0.548$$

Then approximately half propagates toward each end:

$$f_{\text{TIR, end}} \approx 0.274$$

Important distinctions

- $2a - 1$ is for the *rectangular bar* (two wall families simultaneously)
- This is **not** the Knoll slab result (single wall)
- The old $\approx 80\%$ referred to a different quantity/geometry; not directly comparable
- The napkin uses $f_{\text{TIR, end}} \approx 0.274$

Coverage per end: $8 \times 6 \times 6 / (60 \times 10) \approx 0.48$ — geometric scale estimate only; real acceptance differs.

Group velocity:

$$v_g = c/n \approx 300/1.58 \approx 190 \text{ mm/ns}$$

Direct transit time (half-bar from centre):

$$t_{\text{direct}} \approx 700 \text{ mm}/190 \text{ mm/ns} \approx 3.7 \text{ ns}$$

Longitudinal position from timing difference

$$\Delta t = t_R - t_L \approx -2x/v_{\text{eff}}$$

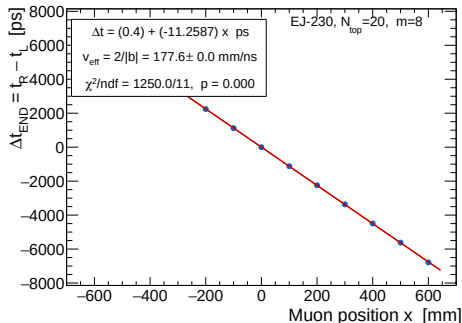
Measuring Δt gives x — the standard END position estimator.

Measured effective velocity from Geant4:

$$\Delta t = (0.44) + (-11.259)x \text{ [ps/mm]}$$

$$v_{\text{eff}} = 177.6 \pm 0.0 \text{ mm/ns}$$

(uncertainty from fit; $\chi^2/\text{ndf} = 1250/11$ — mild non-linearity)



Mean $\Delta t = t_R - t_L$ vs muon position x . Linear fit gives $v_{\text{eff}} = 177.6 \text{ mm/ns}$, lower than $c/n = 190 \text{ mm/ns}$ due to path-lengthening from reflections.

Reflection counting (geometric estimate)

For a photon at angle α from bar axis, transverse bounce in D :

$$N_{\text{bounce}} \approx \frac{d \tan \alpha}{D}$$

- $D = 10$ mm (thickness): $N \approx 86$ for $d = 700$ mm
- $D = 60$ mm (width): $N \approx 14$

With Vikuiti reflectivity $R = 0.98$ (napkin value):

$$P_{\text{surv}} = R^N = e^{N \ln R} = e^{-d/\Lambda_{\text{refl}}}$$

$$\Lambda_{\text{refl}} = \frac{-D}{\tan \alpha \ln R}$$

Λ_{refl} is geometry-dependent

It is **not** a material constant of Vikuiti. It depends on: reflectivity, bar dimensions, photon direction, and which surface family is involved.

Napkin parameter: $R = 0.98$ is an approximation. The actual Vikuiti reflectivity is wavelength-dependent; Geant4 uses the full spectral table.

Estimate: for typical TIR-guided directions ($\alpha \sim 50^\circ$), the effective attenuation length $\Lambda_{\text{refl}} \sim 500\text{--}1000$ mm.

Two loss mechanisms:

① **Bulk attenuation:** $e^{-d/\lambda_{\text{att}}}$

λ_{att} : 900 mm (EJ-230) to 3800 mm (EJ-200)

Effect over 700 mm: factor 0.46 (EJ-230) to 0.83 (EJ-200)

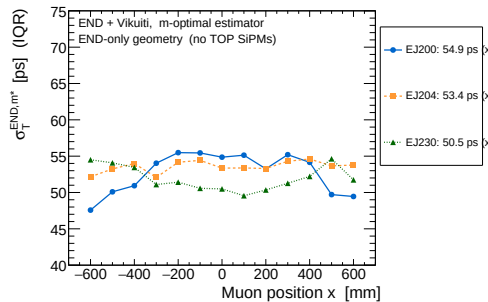
② **Reflector losses:** as above

Timing: dominated by scintillation decay time τ :

$$P_{\text{first}}(t) \propto e^{-t/\tau} \Rightarrow t_{\text{first}} \sim \text{Exp}(\tau)$$

Shorter $\tau \Rightarrow$ earlier first photon $\Rightarrow$ better timing.

Trade-off: EJ-230 ($\tau = 1.5$ ns) wins timing, but EJ-200 ($\tau = 2.1$ ns, $3\times$ more photons) gives better signal-to-noise and is more radiation-hard.



Geant4: σ_{IQR} of T_0^{END} vs muon position x . EJ-230 gives best timing — consistent with napkin prediction. Legend shows σ at $x = 0$.

Napkin chain (EJ-230, per end)

- ① $N_\gamma = 19\,400$
- ② $f_{\text{TIR,end}} = 0.274 \Rightarrow 5312$
- ③ Reflector $P_{\text{surv}} \approx 0.5$ (estimate): $\Rightarrow 2656$
- ④ Coverage 0.48: $\Rightarrow 1275$
- ⑤ $\text{PDE}_{\text{napkin}} = 0.40 \Rightarrow \approx 510 \text{ pe/end}$

Geant4 output (EJ-230, END-only geometry, $x = 0$):

$$N_{\text{pe}}^{\text{END}} \approx 701 \text{ pe/end}$$

PDE audit

The simulation records the PDE applied to each photon.

- Mean effective PDE $\approx$ **0.607**
- NOT 0.40 as in the original napkin
- The AFBR-S4N66P024M peak PDE > 0.60

With PDE = 0.607:

$$N_{\text{pe}}^{\text{napkin}} \approx 1275 \times 0.607 \approx 774 \text{ pe/end}$$

Agreement with Geant4 at $\sim 10\%$ level.

Note: TOP geometry gives $N_{\text{pe}}^{\text{TOP}} \approx 1341 \text{ pe/event}$ for $N_{\text{TOP}} = 20$ SiPMs. This counts all TOP faces — a different optical path.

Two observables from END readout:

Event time (T0):

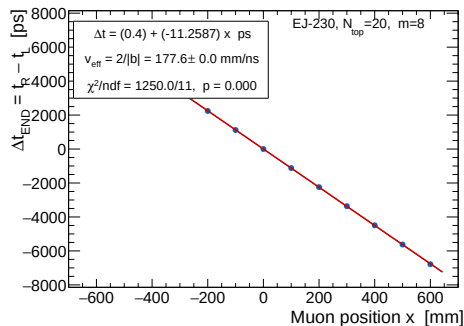
$$T_0^{\text{END}} = \frac{1}{2} \left[\frac{1}{m} \sum_{i=1}^m t_L^{(i)} + \frac{1}{m} \sum_{i=1}^m t_R^{(i)} \right]$$

where $t_L^{(i)}, t_R^{(i)}$ are sorted arrival times at LEFT/RIGHT faces.

Longitudinal position (Δt):

$$\Delta t_{\text{END}} = \frac{1}{m} \sum_{i=1}^m t_R^{(i)} - \frac{1}{m} \sum_{i=1}^m t_L^{(i)}$$

$$x \approx -\frac{v_{\text{eff}}}{2} \Delta t_{\text{END}}$$



Measured $v_{\text{eff}} = 177.6 \text{ mm/ns}$ from linear fit to $\langle \Delta t \rangle$ vs x .

END position resolution

At $m = 8$ (timing optimal): $\sigma_x^{\text{END}} \approx 7.9 \text{ mm}$

At $m = 26$ (position optimal): $\sigma_x^{\text{END}} \approx 7.2 \text{ mm}$

Optimal m for timing and position differ slightly.

Transport is understood.

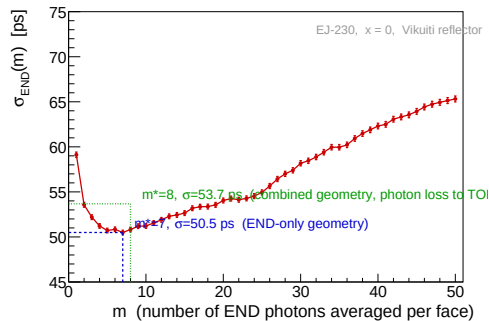
Are we extracting the timing information optimally?

Definition of $T_0^{\text{END}}(m)$: average of first m photons per side.

Two important configurations:

- **END-only** geometry: $m^* = 7$, $\sigma_{\text{IQR}} = 50.5$ ps
- **Hybrid** (TOP SiPMs present): $m^* = 8$, $\sigma_{\text{IQR}} = 53.7$ ps

The hybrid degrades by ~ 3 ps because TOP SiPMs intercept photons that would otherwise reach END faces. **Always use $m^* = 8$ from hybrid geometry** when discussing the combined detector.



σ_{IQR} vs m for EJ-230 at $x = 0$. Minimum at $m^* = 7/50.5$ ps (END-only, dashed) and $m^* = 8/53.7$ ps (hybrid, solid). Both share the same events and material; only the optical boundary changes.

Sparse TOP layout ($N_{\text{TOP}} = 20$):

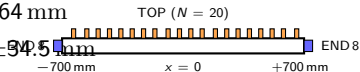
- Sensors at $x = -664, -594, \dots, +594, +664$ mm
- Spacing ≈ 70 mm; nearest to bar centre: ± 34.5 mm
- $N_{\text{TOP}} = 20$ is NOT the total; hybrid total is $N_{\text{total}} = 36$

Pooled estimator: collect all photon arrivals from ALL N_{TOP} sensors into one list, then take the k -th earliest:

$$T_0^{\text{TOP}} = t_{(k)}^{\text{pooled}}$$

This is fundamentally different from per-SiPM averaging.

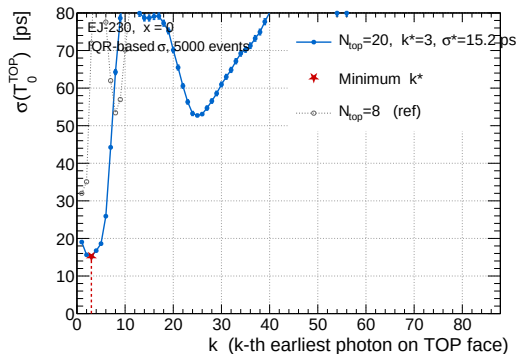
SiPM positions along bar



Orange: 20 TOP SiPMs. Blue: 8 END SiPMs/side.

$N_{\text{total}} = 36/\text{bar}$.

TOP k -scan (EJ-230, $N_{\text{TOP}} = 20$, $x = 0$):



σ_{IQR} vs k . Minimum at $k^* = 3$, $\sigma = 15.2$ ps.

Optimum and interpretation:

- $k^* = 3 \Rightarrow \sigma_{\text{IQR}}(T_{\text{TOP}}) = 15.2$ ps
- $k = 1$ gives 19.0 ps — not optimal
- $k > 3$ averages over photons from more distant sensors
- $k^* = 1$ is not always optimal: the first photon may arrive from a distant SiPM and carry little timing information about the true T_0

Train/test validation:

- Training (2500 ev): $k^* = 3$, $\sigma = 15.6$ ps
- Test (2500 ev): $\sigma = 14.99 \pm 0.38$ ps
- No evidence of over-fitting; global estimator is stable

Why did TOP improve from ~ 70 ps to 15 ps?

Historical legacy result:

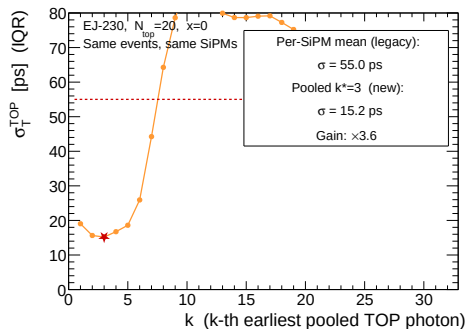
EJ-230 TIR, $N = 70$ SiPMs, per-SiPM mean estimator:

$$\sigma_{\text{TOP}}^{\text{old}} \approx 70 \text{ ps}$$

Per-SiPM mean = average of first-photon arrival per SiPM, then average over SiPMs.

Same events, same $N = 20$ SiPMs, $x = 0$:

- Per-SiPM mean estimator: $\sigma = 55$ ps
- Pooled $k = 1$: $\sigma = 19.0$ ps
- Pooled $k^* = 3$: $\sigma = 15.2$ ps
- **Improvement:** $\times 3.6$



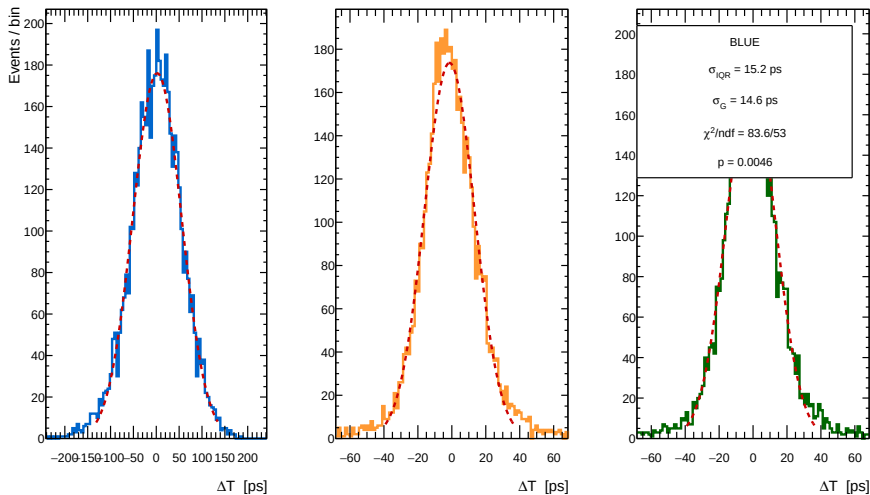
Same events, same SiPMs, $x = 0$, EJ-230 $N = 20$.
Dashed line: per-SiPM mean (55 ps). Curve: pooled k -th order statistic.

Conclusion

The 70 ps $\rightarrow$ 15 ps improvement is primarily an **estimator advance**: pooled k -th vs per-SiPM mean (same events, same SiPMs).

Timing distributions: END nearly Gaussian, TOP/BLUE have tails

EJ-230, $N_{\text{top}}=20$, $x=0$, 5000 events $\hat{\sigma}_{\text{IQR}} = (Q75-Q25)/1.349$



$\sigma_{\text{IQR}} = (Q75 - Q25)/1.349$. **END:** 53.7 ps ($p = 0.09$). **TOP:** 15.2 ps ($p = 0.19$). **BLUE:** 15.2 ps ($p = 0.005$, non-Gaussian $\Rightarrow$ use σ_{IQR}).

Define two calibrated estimators T_{END} , T_{TOP} for the same event time T_0 .

$$T_{\text{BLUE}} = w_{\text{END}} T_{\text{END}} + w_{\text{TOP}} T_{\text{TOP}}, \quad w_{\text{END}} + w_{\text{TOP}} = 1$$

Covariance matrix:

$$C = \begin{pmatrix} \sigma_E^2 & \text{Cov} \\ \text{Cov} & \sigma_T^2 \end{pmatrix}$$

BLUE weights (minimum variance, linear, unbiased):

$$\vec{w} = \frac{C^{-1} \mathbf{1}}{\mathbf{1}^T C^{-1} \mathbf{1}}, \quad \sigma_{\text{BLUE}}^2 = \frac{1}{\mathbf{1}^T C^{-1} \mathbf{1}}$$

$$w_{\text{END}} = \frac{\sigma_T^2 - C_{ET}}{\sigma_E^2 + \sigma_T^2 - 2C_{ET}}$$

Physical interpretation: BLUE weights each estimator by its precision *after* accounting for correlations.

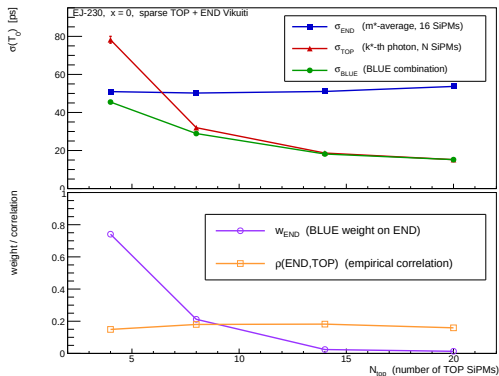
At $N_{\text{TOP}} = 20$, $x = 0$:

- $\sigma_E = 53.7$ ps, $\sigma_T = 15.2$ ps
- $\rho_{\text{ET}} = 0.16$ (non-negligible)
- $w_{\text{END}} = 0.013$, $w_{\text{TOP}} = 0.987$
- $\sigma_{\text{IQR}}(T_{\text{BLUE}}) = 15.2$ ps

Key message

At $N_{\text{TOP}} = 20$: $w_{\text{END}} \approx 1\%$,
 $\sigma_{\text{BLUE}} \approx \sigma_{\text{TOP}} \approx 15$ ps.

N_{TOP} scan: BLUE collapses onto TOP

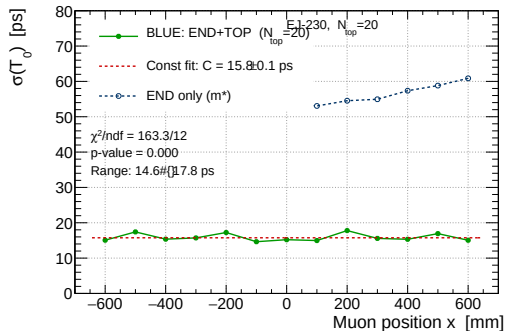


Top: σ_{IQR} vs N_{TOP} for EJ-230, $x = 0$. Bottom: w_{END} and ρ_{ET} vs N_{TOP} .

N_T	σ_E [ps]	σ_T [ps]	ρ	w_E	σ_B [ps]
4	51.0	78.2	.15	.74	45.5
8	50.2	32.0	.18	.21	28.9
14	51.0	18.7	.18	.02	18.2
20	53.7	15.2	.16	.01	15.2

- At $N_{\text{TOP}} = 4$: END contributes 74%; BLUE genuinely combines both
- At $N_{\text{TOP}} = 14$: END contributes $< 2\%$
- At $N_{\text{TOP}} = 20$: BLUE $\equiv$ TOP

This is a *success* of BLUE, not a failure: the estimator correctly identifies which measurement is more precise.



σ_{IQR} vs muon position x for EJ-230, $N_{\text{TOP}} = 20$. BLUE (green, error bars) and END alone (blue dashed). Constant fit to BLUE shown; χ^2/ndf tests uniformity.

Summary:

- $\sigma_{\text{IQR}}(T_{\text{BLUE}})$ ranges 15–23 ps (global $k^* = 3$, $m^* = 8$)
- Peak-to-peak variation $\sim 50\%$
- Positions between TOP sensors ($x = \pm 200$, ± 500) give $\sigma \approx 23$ ps
- Near-sensor positions give $\sigma \approx 15$ ps

TOP timing bias (relative to $x = 0$):

- END bias: < 15 ps, nearly flat
- TOP bias: 0 to -104 ps, position-dependent
- Discrete levels (3 groups by nearest-SiPM distance)
- TOP needs **position-dependent calibration**

[See backup: figE_bias_vs_x]

END timing resolution (hybrid geometry):

$$\sigma_{\text{IQR}}(T_{\text{END}}) \approx 53.7 \pm 0.9 \text{ ps} \quad (m^* = 8)$$

Weight in BLUE at $N_{\text{TOP}} = 20$: $w_{\text{END}} \approx 1\%$.

END is nearly useless for event time at $N_{\text{TOP}} = 20$.

END position resolution:

$$\sigma_x^{\text{END}} \approx 7.9 \text{ mm} \quad (m = 8) \quad \text{or} \quad 7.2 \text{ mm} \quad (m = 26)$$

TOP centroid position (N_{pe} -weighted):

- Bias-calibrated: $\sigma_x \approx 8\text{--}11 \text{ mm}$
- Comparable to END (7–8 mm)
- Requires offline calibration map

Key finding

At $N_{\text{TOP}} = 20$, TOP alone provides:

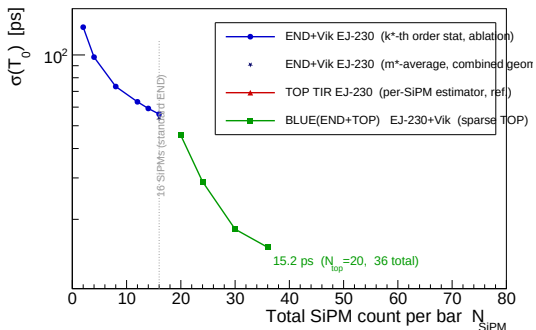
- 1 Timing: $\sigma_t \approx 15 \text{ ps}$
- 2 Position: $\sigma_x \approx 8\text{--}11 \text{ mm}$ (calibrated centroid)

END provides comparable position ($\sigma_x \approx 7\text{--}8 \text{ mm}$) but negligible timing contribution.

Design question: if TOP already provides timing AND position, can we remove END?

Answer requires: a Geant4 simulation of the TOP-only physical geometry (END faces replaced with optical absorber/reflector). This changes the photon boundary condition and cannot be inferred from software masking alone.

[TOP-only run: proposed as next step]



Timing resolution vs total SiPM channels per bar. Black: END-only (8 SiPMs/end $\times$ 2, N_{total} as shown). Star:

m -average point at $N = 16$. Green: BLUE hybrid (16 END + N_{TOP} TOP, $N_{total} = 16 + N_{TOP}$).

Key points:

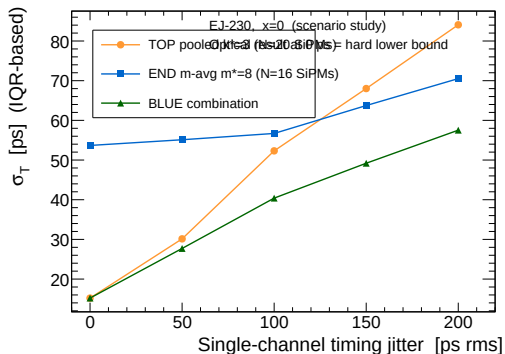
- END-only ($N = 16$, $m^* = 8$): $\sigma \approx 53.7$ ps
- BLUE $N_{TOP} = 4$ ($N_{tot} = 20$): $\sigma \approx 45.5$ ps
- BLUE $N_{TOP} = 8$ ($N_{tot} = 24$): $\sigma \approx 28.9$ ps
- BLUE $N_{TOP} = 14$ ($N_{tot} = 30$): $\sigma \approx 18.2$ ps
- **BLUE $N_{TOP} = 20$ ($N_{tot} = 36$): $\sigma \approx 15.2$ ps**

Adding 20 sparse TOP sensors to 16 END sensors:

- $\times 3.5$ improvement in timing
- $\times 2.25$ increase in channel count

Note: all END channels use pooled k -th estimator with best-available k per geometry; all hybrid uses BLUE with global k^* , m^* .

Electronics jitter: how far does 15 ps survive?



σ_{IQR} vs single-channel timing jitter (SPTR, rms). EJ-230, $x = 0$, $N_{\text{TOP}} = 20$. Independent Gaussian jitter per photon arrival.

Jitter scenarios (optical limit at 0 ps):

Jitter [ps]	σ_{TOP} [ps]	σ_{END} [ps]	σ_{BLUE} [ps]
0	15.2	53.7	15.2
50	30.1	55.1	27.7
100	52.3	56.7	40.4
150	68.0	63.7	49.2
200	84.1	70.5	57.5

Key observations:

- At 50 ps jitter: TOP $\rightarrow$ 30 ps (factor 2 degradation)
- At ≥ 150 ps: END becomes competitive with TOP
- The $k = 3$ pooled estimator amplifies per-photon jitter
- Realistic jitter target: < 50 ps SPTR for < 30 ps event time

What we know from this analysis:

- 1 TOP ($N_{\text{TOP}} = 20$) gives $\sigma_t \approx 15$ ps and $\sigma_x \approx 9$ mm
- 2 END adds $< 1.3\%$ weight to event time at $N_{\text{TOP}} = 20$
- 3 END gives $\sigma_x \approx 7\text{--}8$ mm — comparable to TOP
- 4 TOP has strong position-dependent timing bias; requires calibration
- 5 At ≥ 50 ps channel jitter, TOP degrades significantly
- 6 Uniformity: σ_{BLUE} varies 15–23 ps with position (50% peak-peak)

Open questions:

- **TOP-only geometry:** does removing END physically change the 15 ps result? (boundary conditions change; needs new Geant4 run)
- **Optimal TOP layout:** uniform spacing may not minimise worst-case $\sigma_t(x)$
- **Common clock jitter:** separate from per-channel SPTR; needs detector-level study
- **TOP-alone position:** 8–11 mm adequate? Or do we need END for < 8 mm?

Current recommendation

Keep $N_{\text{TOP}} = 20$ hybrid pending TOP-only Geant4 result. If TOP-only gives equivalent timing, 16 END channels per bar ($\times 600$ bars = 9600 channels) can be eliminated.

❶ First principles explain the optical transport

TIR geometry, Vikuiti reflector losses, and scintillation kinetics correctly predict the scale and material ordering of Geant4. The effective PDE in simulation (≈ 0.61) is higher than the original napkin assumption (0.40); the updated napkin gives $\sim 10\%$ agreement with Geant4.

❷ The estimator choice dominates timing performance

A pooled k -th order statistic on TOP outperforms the legacy per-SiPM mean estimator by a factor $\times 3.6$ on the same events. The $k^* = 1$ first-photon is not optimal; $k^* = 3$ gives 15.2 ps.

❸ At $N_{\text{TOP}} = 20$, TOP dominates event time; BLUE $\equiv$ TOP

BLUE assigns $< 1.3\%$ weight to END ($w_{\text{END}} = 0.013$). Within statistical uncertainty, $\sigma_{\text{IQR}}(T_{\text{BLUE}}) \approx \sigma_{\text{IQR}}(T_{\text{TOP}}) \approx 15$ ps. Global (train/test) and oracle estimators agree to < 0.4 ps.

❹ The remaining problem is detector-level

Position-dependent TOP timing bias (≤ 104 ps) requires calibration. Uniformity is $\pm 50\%$ peak-to-peak along the bar. Electronics jitter $\gtrsim 50$ ps degrades TOP to > 30 ps. The value of END depends on the TOP-only physical geometry — a targeted Geant4 run is the critical next step.

Backup slides

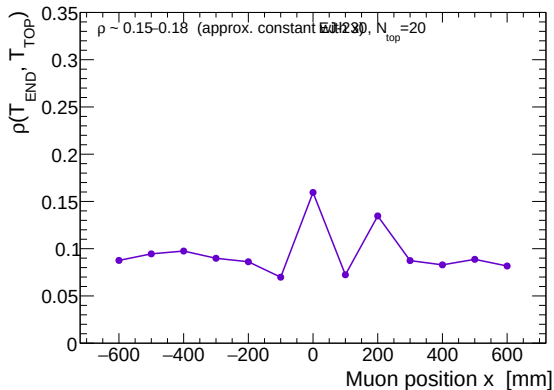
Methodology (to avoid optimistic bias):

- ① Split 5000 events 50/50 by random permutation (seed 20260820)
- ② Training half: find k^*, m^* by minimising σ_{IQR}
- ③ Test half: apply global k^*, m^* without re-optimising

Results at $x = 0$, EJ-230, $N_{\text{TOP}} = 20$:

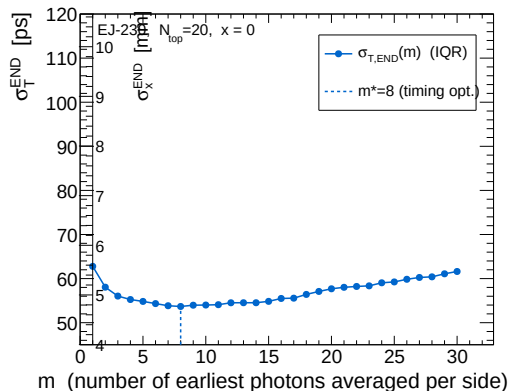
Mode	k^*	σ_{BLUE}
Oracle (same-sample)	3	15.21 ps
Global (train/test)	3	14.85 ± 0.40 ps
Difference	—	0.36 ps

The oracle optimistic bias is < 0.4 ps, well within bootstrap uncertainty. Both oracle and global agree on $k^* = 3$.

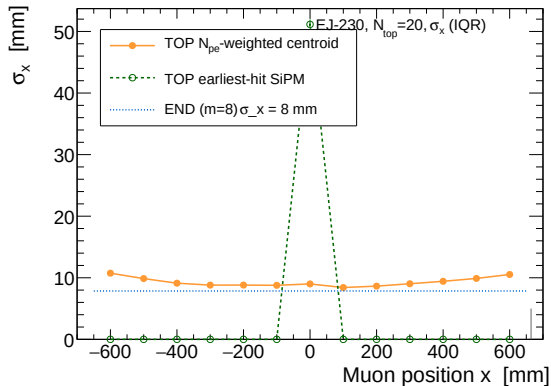
**Key findings:**

- ρ_{ET} varies from 0.07 to 0.16 across the bar
- Not constant with x — correlation depends on the local photon geometry
- w_{END} follows $\sigma_T(x)$: where TOP is worse, END gets more weight
- At positions between TOP sensors (high σ_T), w_{END} increases to $\sim 10\%$

The BLUE weights are computed position-by-position from training data.



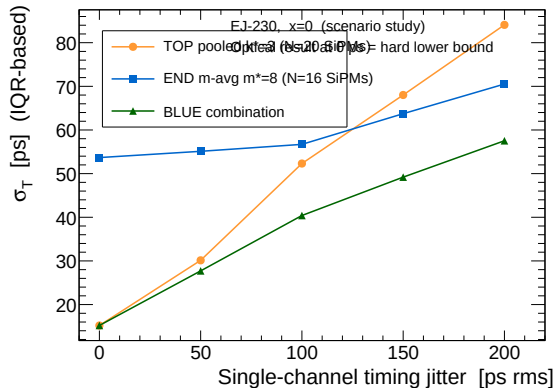
- Left: $\sigma_{\text{IQR}}(T_{\text{END}})$ vs m (timing)
- Right: $\sigma_x^{\text{END}} = (v_{\text{eff}}/2)\sigma_{\text{IQR}}(\Delta t_m)$
- Timing opt: $m^* = 8$, $\sigma_T = 53.7$ ps, $\sigma_x = 7.9$ mm
- Position opt: $m = 26$, $\sigma_x = 7.2$ mm
- Difference 0.7 mm: use $m = 8$ for both



Simple TOP position estimators:

- **Earliest-hit SiPM:** discrete, bias ≤ 25 mm
- **N_{pe} centroid:** $\sigma_x \approx 8\text{--}11$ mm (calibrated)
- Uncorrected centroid bias ≤ 300 mm (centre pull)
- Cal: $x_{est} = f(\text{centroid})$ from training data

Horizontal dashes: END position resolution at $m = m^* = 8$ ($\sigma_x \approx 7.9$ mm). END and TOP centroid give comparable position resolution.



Jitter model:

- Gaussian SPTR per photon, independent
- $t_i \rightarrow t_i + \mathcal{N}(0, \sigma_j^2)$, re-sort
- Scenario study (not measured)

Physical interpretation:

- TOP: k -th photon displaced by jitter
- END m -avg: jitter reduced by $\sqrt{m}$
- $\sigma_j = 150$ ps: END $\approx$ TOP

Simulation PDE (AFBR-S4N66P024M, from data):

- Mean PDE (all detected photons): 0.607
- Range: 0.00–0.63 (wavelength-dependent)
- No significant difference between END and TOP faces

Updated napkin (EJ-230, per end, PDE= 0.61):

$$N_{\text{pe}}^{\text{napkin}} \approx 1275 \times 0.61 \approx 778 \text{ pe/end}$$

vs Geant4: $\approx 701 \text{ pe/end}$.

Agreement at $\sim 10\%$ — better than with PDE= 0.40 ($\sim 40\%$ low).

Corrected photon-budget comparison (per end)

Quantity	Napkin	Geant4
N_{γ} produced	19 400	—
$f_{\text{TIR,end}}$	0.274	—
$P_{\text{surv}} \times \text{coverage}$	$\sim 0.50 \times 0.48$	—
PDE	0.61	0.61 (measured)
N_{pe}/end	≈ 778	≈ 701
Agreement	$\approx 10\%$	

Motivation: at $N_{\text{TOP}} = 20$, BLUE gives $w_{\text{END}} \approx 1\%$. Can the 16 END channels be eliminated?

What cannot be inferred from existing data:

- Removing END SiPMs changes the optical boundary condition at the end faces
- Photons that currently enter END SiPMs would instead be reflected/absorbed
- This modifies the photon population reaching TOP SiPMs
- The effect on σ_t^{TOP} is unknown without a physical simulation

Proposed run:

- Same bar, same Vikuiti wrapping, same $N_{\text{TOP}} = 20$ positions
- End faces: replace SiPMs with physical reflector or absorber
- Same muon scan (all 13 $\times$ positions), 5000 events each
- Apply same pooled- k estimator; compare with hybrid result

Expected outcome: if TOP-only gives $\sigma_t \approx 15$ ps and $\sigma_x \approx 10$ mm, then 16 END channels/bar ($\times 600$ bars = 9600 channels) can be eliminated from the full detector.